



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 17-400-2

Date: 29-09-2020

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TECHNICAL SPECIFICATION

FOR

MEDIUM VOLTAGE CURRENT TRANSFORMERS



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Issue: SEP-2020 / Rev- 2

❖ توجد بنود اختيارية Option items يجب تحديدها بواسطة شركة التوزيع قبل الطرح.



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Scope:

This specification covers the minimum technical requirements for design, manufacturing, testing, packing and procurement of medium voltage current transformers used for measuring and protective relays at 50HZ.

General Description:

Insulated Current Transformers (CTs) are casted in epoxy resin and designed for rated voltages up to 24 kV. CTs are fabricated with required dimensions, single ratio or double ratios having the possibility to be re-connectable on the secondary side according toEDC requirements.

Applicable standards:

Unless otherwise specified, the current transformer should comply with the latest edition of IEC publications or any other equivalent international standard specification approved by (....EDC).

- IEC 61869-1, Instrument transformers general requirements
- IEC 61869-2, Additional requirements for current transformers.
- And any relevant IEC standards

Environmental conditions:

The performance of the current transformer should be guaranteed for the following table, any differences in the guaranteed performance should be clearly set out in the offer:

Minimum ambient temperature	-5°C
Maximum ambient temperature	45°C (50 °c as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m.



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Technical specifications:

- Highest voltage for equipment U_m (R.M.S) (kV) 12 – 24
- Number of secondary windings 1-2 (3 if required)
- Service voltage (KV) 11 - 22
- Secondary current (A) 5 (1 if required)
- Rated primary current (A) **As Required**
- Rated continuous current 1.2 I rated primary current
- Accuracy class :
 - Measuring 0.5S
 - Protection 5P10- 5P20
- Rated burden:
 - Measuring (VA) ≥ 10
 - Protection (VA) ≥ 15
- Accuracy Limiting Factor
 - Protection P10 – P20
 - Measuring FS5 – FS10
- Creepage distance ≥ 2 Cm/KV
- Rated short time "thermal" withstand current (RMS) (KA)

	1 sec winding 11KV	1 sec winding 22 KV	2 sec windings 11 KV	2 sec windings 22 KV
Time	1 sec	1 sec	3 sec	3 sec
Value of SC (KA)	25	20	31.5or 25 ⁽¹⁾	25

⁽¹⁾ Please choose only one value.

- Rated dynamic current I_p (peak) 2.5x I thermal
- Insulation class $\geq$ E (temperature rise limit "K" 70 °C @ 45°C ambient (65@50°C ambient)
- Rated impulse withstands voltage for primary windings 75 KV for 11KV and 125 KV for 22 KV.
- Short duration withstand voltage (r.m. s) for primary windings (1min.) 28 KV for 12 KV and 50KV for 22 KV.
- Rated power frequency short duration withstands voltage for secondary winding (1 min.) 3 KV.



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DESIGN CRITERIA:

- The current transformer should be designed according to latest edition of IEC standard publication IEC 61869-1, IEC 61869-2.
- The primary current, secondary current and voltage ratings should be as specified above.
- Short time current ratings of protective and metering current transformers should be equal to or higher than that of associated switchgear and equipment as applicable and should meet the above requirements.
- Continuous thermal current for all CTs at highest tap should be as per above requirements without exceeding temperature limits.
- Unless otherwise specified all current transformers CT core for protection should have ratings and performance requirements of relaying including transient performance characteristics at specified tap.
- CTs should be designed to operate continuously for the respective application /installations.
- Unless otherwise specified, the metering/protection class should be as per IEC at specified tap.
- The current Tr. shall have terminal box for the secondary terminal and earth terminal.
- The routine test certificate including magnetization curve for the core material should be submitted.
- The current Tr. Should be interchangeable from lower ratio to higher ratio from the secondary side of the current Tr.
- Maximum winding temperature rise should be according to IEC 61869-2.
- Limits of current error and phase displacement should be in accordance with IEC 61869-2.



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Type tests:

- Temperature-rise test
- Impulse voltage withstand test on primary terminals
- Tests for accuracy
- Short-time current tests

Type tests should be in accordance with IEC 61869-1,2.

Type test certificates should be submitted with the offer.

Routine/Acceptance Tests:

- Power-frequency voltage withstand tests on primary terminals.
- Partial discharge measurement
- Power-frequency voltage withstand tests between sections
- Power-frequency voltage withstand tests on secondary terminals
- Tests for accuracy
- Verification of markings
- Test for rated knee point E.M.F and exciting current at rated knee point E.M.F.(for PX class)
- Inter-turn overvoltage test

Acceptance tests should be performed on random samples and should be selected and witnessed by Distribution Company representative.

Routine tests should be performed in compliance with IEC61869-1,2

Marking:

The following data should be marked on CT itself or on a nameplate engraved or laser printed on the CT, and located in a place such that they are visible and legible when CT is installed:

- The Manufacturer's Name/ model;
- Serial Number and Year of Manufacturing
- Applicable C.T. Standard
- Rated Frequency
- Current Transformer Ratio
- Measuring Core Burden and Accuracy
- Protection Core Burden and Accuracy
- 3rd core burden and accuracy(if 3rd core is required)
- Rated Insulation level & highest voltage of equipment.



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- Power Frequency Withstand Voltage(Kv)
- Impulse Withstand Voltage(Kv)
- Short Time Withstand Current (I_{sc} in kA for 1 or 3 sec)
- Rated dynamic current ($I_{dyn.} = 2.5 I_{sc}$)
- Class of insulation
- On current transformers with two or more secondary windings, the use of each winding and its corresponding terminals.

All C.T. should carry at least the clear and indelible markings as specified in IEC 61869-1,2.

Drawings and Catalogues: -

The tenderer should submit a complete set of dimensional drawings, technical catalogues and user manual showing the CT type offered and its performance in full details.

Sample: -

The tenderer should provide a sample C.T. with his offer (if Distribution Company required). The sample should correspond in all respects with the material they intend to supply.

Guarantee Table:

Tenders not accompanied with attached table of guarantee fulfilled and signed by tenderers shall not be considered.

Warranty Period:

Warranty period shall be 18 months from delivery or 12 months from operation date which later. For the first submittal, the warranty period should be extended to be 3 years at least ended with a performance report prepared by correspondingEDC. This report should prove product high quality.

ملاحظة:

يلزم تقديم (ضمن مستندات العرض) نسخة من شهادة إختبار مركز أبحاث الجهد الفائق تفيد اجتياز محولات التيار من نفس الطراز ولنفس القيم المقننة للاختبارات النوعية التي أجريت عليها و تستكمل باقي الإختبارات النوعية من معمل معتمد ومحاييد طبقا للمواصفات القياسية IEC علي ان يتم الإطلاع علي أصل أو صورة طبق الأصل من شهادات الإختبارات النوعية.



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GUARANTEE TABLE

- Applicable Standard/s	
- Name of Manufacturer	
- Place of manufacturing	
- Year of manufacturing	
- Product type	
- Highest voltage for equipment U_m (R.M.S)	KV
- Rated primary service voltage	KV
- Rated frequency	Hz
- Rated primary current	A
- Rated secondary current	A
- Winding 1	
- Winding 2	
- Winding 3(if required)	
- Current continuous factor	
- Rated short time "thermal" withstand current (RMS) For 1 sec (KA)	
- Rated short time "thermal" withstand current (RMS) For 3 sec (KA)	
- Rated dynamic current I_{dyn} (KA)	
- Ambient max. Temperature	°C
- Rated impulse withstand voltage for primary windings	KV
- Short duration withstand (R.M.S) voltage for primary windings (1min)	KV
- Rated power frequency short duration withstand for secondary winding..... For 1 minute	KV
- Rated Burden	VA
- Winding 1	
- Winding 2	
- Winding 3(if required)	
- Accuracy class	
- Winding 1	
- Winding 2	
- Winding 3(if required)	
- Creepage distance	Cm
- Class of insulation	
- Marking:	
- Marking of the rating plate should be according to IEC 61869-1,2	(Yes / no)